

CS532 HW4

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Problem 7

Problem

Consider the network fragment shown below. x has only two attached neighbors, w and y . w has a minimum-cost path to destination u (not shown) of 5, and y has a minimum-cost path to u of 6. The complete paths from w and y to u (and between w and y) are not shown. All link costs in the network have strictly positive integer values.

1. Give x 's distance vector for destinations w , y , and u .
2. Give a link-cost change for either $c(x, w)$ or $c(x, y)$ such that x will inform its neighbors of a new minimum-cost path to u as a result of executing the distance-vector algorithm.
3. Give a link-cost change for either $c(x, w)$ or $c(x, y)$ such that x will not inform its neighbors of a new minimum-cost path to u as a result of executing the distance-vector algorithm.

Solution

1. $D_x(w) = 2$; $D_x(y) = 4$; $D_x(u) = 7$
2. $c_{x,w} = 7$, then $D_x(w) = 7$; $D_x(y) = 5$ and $D_x(u)$ gets updated as

$$\begin{aligned} D'_x(u) &= \min(c_{x,y} + D_y(u), c_{x,w} + D_w(u)) \\ &= \min(5 + 6, 7 + 5) \\ &= \min(11, 12) \\ D'_x(u) &= 11 \neq 7 \end{aligned}$$

3. $c_{x,y} = 6$, then $D_x(w) = 2$; $D_x(y) = 6$ and $D_x(u)$ gets updated as

$$\begin{aligned} D'_x(u) &= \min(c_{x,y} + D_y(u), c_{x,w} + D_w(u)) \\ &= \min(6 + 6, 2 + 5) \\ &= \min(12, 7) \\ D'_x(u) &= 7 \end{aligned}$$

Problem 14

Problem

Consider the network shown below. Suppose AS3 and AS2 are running OSPF for their intra-AS routing protocol. Suppose AS1 and AS4 are running RIP for their intra-AS routing protocol. Suppose eBGP and iBGP are used for the inter-AS routing protocol. Initially suppose there is no physical link between AS2 and AS4.

1. Router 3c learns about prefix x from which routing protocol: OSPF, RIP, eBGP, or iBGP?
2. Router 3a learns about x from which routing protocol?
3. Router 1c learns about x from which routing protocol?
4. Router 1d learns about x from which routing protocol?

Solution

1. eBGP: 3c talks to 4c to get reachability and prefix x.
2. iBGP: 3c communicates to 3a in its AS.
3. eBGP: 1c talks to 3c which has the prefix x.
4. iBGP: 1c talks to 1d in its AS.

Problem 16

Problem

Consider the following network. ISP B provides national backbone service to regional ISP A. ISP C provides national backbone service to regional ISP D. Each ISP consists of one AS. B and C peer with each other in two places using BGP. Consider traffic going from A to D. B would prefer to hand that traffic over to C on the West Coast (so that C would have to absorb the cost of carrying the traffic cross-country), while C would prefer to get the traffic via its East Coast peering point with B (so that B would have carried the traffic across the country). What BGP mechanism might C use, so that B would hand over A-to-D traffic at its East Coast peering point? To answer this question, you will need to dig into the BGP specification.

Solution

I think C should use MULTI_EXIT_DISC. Here is what the docs say about such:

The MULTI_EXIT_DISC is an optional non-transitive attribute that is intended to be used on external (inter-AS) links to discriminate among multiple exit or entry points to the same neighboring AS. The value of the MULTI_EXIT_DISC attribute is a four-octet unsigned number, called a metric. All other factors being equal, the exit point with the lower metric SHOULD be preferred.

Using this, C could advertise a lower MED at the East-Coast peering to prioritize traffic to be routed there. This is assuming that LOCAL_PREF and AS_PATH are equal, but being the case, this would let C receive the traffic at the East-Coast peering.